#uni #s26 #bu419

Lecture 02: Purpose & Operation of Insurance

Purpose of Insurance

- The biggest problem that businesses or individuals face is the uncertainty of future events.
- In any year, a business could face severe catastrophic events that could cause it to completely shut down, which cannot be predicted at the start of the year.
- Therefore, uncertainty brings with it a lot of risk.
- On the other hand, insurers work with several similar businesses and can study historical data and make accurate predictions for the group of businesses they work with. Remember the law of large numbers that was discussed in the previous chapter.
- Insurance therefore allows businesses or individuals to transfer their risk to the insurer in exchange for a premium.
- This allows businesses to transfer the uncertainty and the risk associated with it and the potential financial implications to the insurer.

Pooling

- Pooling is fundamental to the operation of insurance.
- Pooling reduces risk when the pool losses are independent or uncorrelated.
  - Losses are independent when each loss occurs independently and they are not subject to a common cause of loss.
- Let's look at an example of how pooling reduces individual risks for businesses:
  - Consider Organization A and Organization B.
  - The probability of loss for both organizations is 20%, and the amount of the loss is $2,500.
  - Individually, the expected loss for both organizations would be

$$E(\text{loss}) = (0.20 * 2500) + (0.80 * 0) = \$500$$

  - When pooled, the new probabilities are:
    - $P(OrgAloss) = P(OrgBloss) = \$0.20 * 0.80 = 0.16$
    - P(Org A loss & Org B loss) = $0.20 * 0.20 = 0.04$
    - P(No loss) = $1 - 0.16 - 0.16 - 0.04 = 0.64$

    - Now the expected loss for each organization would be

$$E(\text{loss}) = \frac{(0.16 * 2500) + (0.16 * 2500) + (0.04 * 5000) + (0.64 * 0)}{2} = \frac{\$1000}{2} = \$500$$

    - So the expected loss for each organization is still $\$500$. However, the probability of the highest loss reduces from 20% to just 4%.
    - Therefore, pooling has reduced the risk of the highest possible loss. The company will still, on average, have to pay the same amount, but it is more likely to pay less than it would have without pooling.
    - In simple terms, without pooling, the company would pay either 0 or $2,500, whereas with pooling, the company will most likely pay $1,250.
- As the number of pool participants increases, the probability of extreme outcomes decreases. The probability distribution of each participant's loss becomes more bell-shaped.
- Pooling increases the predictability of each of its participants' losses by reducing the variability of the average loss and therefore reduces risk.
- Positively correlated losses increase the probability that multiple pool participants will suffer simultaneous losses. This basically increases the variability of their loss distribution, and therefore average losses are more difficult to predict.

Insurance vs Pooling

- Pooling is a mechanism for sharing losses, but not for transferring risk.
- Insurance is based on loss-sharing principles, but it is a risk-transfer mechanism. Insurance transfers risk from the insured to the insurer in exchange for premiums.
- An insurer has additional financial resources from which it can fund losses, enabling it to provide a stronger guarantee that sufficient funds will be available in the event of loss.

Insurer-Provided Risk Management Services

1. Risk Control Services
    - Identification: The insurer assists the insured in identifying potential loss exposures.
    - Recommendation: They provide concrete recommendations on how to control and mitigate those identified risks.
    - This can be packaged directly as a complement to your existing insurance coverage, or handled as an entirely separate, fee-based service.
2. Claims and Legal Services

- Administration: This includes settling claims, managing the actual claims payments, and actively working to prevent fraud.
- Legal Expertise: Insurers provide immense value here because they keep an in-house team of legal experts and maintain a massive network of specialized legal resources to defend you.

Deductibles

- Deductible is the portion of a covered loss that the insurer does not pay.
- From the insured's perspective, it represents risk retention because you are choosing to keep and pay for a small, manageable piece of the risk yourself.
- From the insurer's perspective, it reduces the size of the claim they have to pay. It subsequently lowers the premium.
- Deductibles keep insurance economical. Insurance is meant to transfer financial consequences that the insured cannot afford to retain.
  - It's not meant to cover all the tiny expenses that cost more to process as a claim than the loss itself.

Types of Property Deductibles

- Flat or Straight Deductible: A fixed dollar amount stated in the policy.
- Disappearing or Franchise Deductible: A unique deductible that actually decreases as the size of your loss increases. If the loss crosses a specified large threshold, the deductible disappears completely.
- Percentage deductible: The deductible is calculated as a percentage of a baseline amount, such as:
  - the total amount of insurance tracking the policy
  - the overall value of the property
  - the specific loss amount
- Aggregate Annual Deductible. This caps the total amount of losses you have to retain yourself over a whole year.

Liability Deductibles

- Insureds are much more likely to retain property losses than liability losses. This is because liability lawsuits involve highly volatile and intangible elements, which could increase the cost, which is why organizations prefer to have more insurance and less retention for liability.

- Liability deductibles are structured in a few ways:
  - Per claim deductible: Applies separately to each person or organization that makes a claim against you.
  - Per accident/occurrence deductible: The deductible applies just once to the entire event, regardless of how many individuals step forward to sue you.
  - Waiting period: A statutory time-based deductible where an injured party must wait a certain amount of time before benefits kick in.
- For massive organizations, retention goes a step further:
  - Large Deductible Plans are insurance policies where the per-occurrence or per-accident deductible is $100,000 or more.
  - Self-insurance is a formal recorded internal system where an organization tracks its own losses and maintains its own dedicated funds to pay for them.
    - Frequently used for workers' compensation and auto liability.
  - Legal defense costs and claims adjusting expenses can be classified as inside the deductible or outside the deductible.
    - If inside, the insurer adds it to the amount of loss.
    - If outside, it's not added to the amount of loss and usually prorated between the insured and the insurer.

Excess or Umbrella Liability Coverage

- The need for excess liability insurance is because of:
  - Difficulty in estimating the maximum possible loss (MPL) for liability loss exposures
    - While property losses have a clear physical limit, liability exposures have no easy ceiling.
    - Court awards can skyrocket into millions or even billions of dollars.
  - The layering of liability coverages
    - Primary insurers are often unwilling to provide coverage beyond a standard limit.
    - To secure higher limits, organizations must purchase additional policies stacked on top of each other.
  - Effect of Aggregate Limits
    - Most policies have an overall aggregate limit, which is the maximum amount the policy will pay for all claims combined during a policy period.
    - If a business hits that cap, the primary policy will pay nothing for subsequent losses, leaving the company completely exposed.
- Excess Liability Policies:
  - Designed to provide excess limits of coverage above the limits of the underlying coverage.

    - Offers no broader protection and may be even more restrictive than the underlying policy.
    - Basically, this is just a way to add more financial runway to the exact same scenarios covered by the base policy.
- Umbrella Liability Policy
    - A type of excess liability policy that provides additional limits but also provides coverage not available on underlying coverage and provides defence coverage.
    - The full limit of your umbrella policy is available to pay for damages in the wider domain or for legal defence.
    - You don't just get a portion of the money for these new scenarios.

Structuring a Liability Insurance Program

- Liability insurance is often arranged in layers.
    - Primary Layer
        - Consists of one or more primary coverages that protect against specific standard everyday risks.
        - Examples: CGL, Business Auto, Employers' Liability
    - Working Layer
        - This refers to the layer or layers within the insurance program that are most often called on to pay claims.
        - Because these layers take the first hit, they experience the highest frequency of claims activity.
    - Buffer Layer
        - It is a level of excess insurance coverage that sits directly between a primary layer and an umbrella policy.
        - It is needed because an umbrella insurer might refuse to issue a policy unless you already have a high amount of underlying primary coverage.
    - Umbrella and Excess Layer
        - Refers to the umbrella or excess policies that organizations may have.